Evaluating Geospatial Foundation Models for Irrigated Cropland Mapping in North Africa and Their Transferability to Data-Scarce Regions

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1. Introduction

1.1 Societal and Scientific Relevance

Irrigated agriculture plays a critical role in global food security and water resource management. Although irrigated cropland covers only approximately 20% of the total cultivated area worldwide, it accounts for around 40% of global food production and consumes approximately 70% of all freshwater withdrawals (Wu et al., 2023). As population growth, climate change, and increasing water scarcity place greater pressure on agricultural systems, reliable information on the spatial distribution and temporal dynamics of irrigation has become essential for sustainable resource management, agricultural planning, and progress toward the United Nations Sustainable Development Goals, particularly SDG 2 (Zero Hunger) and SDG 6 (Clean Water and Sanitation).

Morocco provides a suitable case study for investigating irrigated cropland mapping due to its extensive irrigation infrastructure, diverse agricultural systems, and increasing pressure on water resources. Agriculture contributes approximately 14% of Morocco's gross domestic product, accounts for 39% of national employment and 74% of rural employment, and the country has developed an irrigated area of over two million hectares across seven large irrigation schemes (Guemouria et al., 2023). Yet Morocco is also ranked among the most water-stressed countries in the world, with the agricultural sector consuming over 80% of available water resources while rainfall has declined by approximately 30% over the past three decades (Guemouria et al., 2023). The government's response - the Green Morocco Plan (2008-2020) and its successor the Green Generation strategy (2020-2030) - has driven a major expansion of drip irrigation, growing from 160,000 ha in 2008 to 585,000 ha by 2019 (Guemouria et al., 2023). Tracking whether this rapid irrigation expansion is achieving its intended outcomes, and where water use is most intensive, requires precise and up-to-date irrigated cropland maps. Such maps are also essential inputs for evaluating the impact of public policies on agricultural water use at regional scales (Ouassanouan et al., 2026).

Morocco's relatively rich data environment stands in contrast to much of North Africa, where limited irrigation development and severe data scarcity present a different set of challenges. Tunisia is a compelling example: despite sharing Morocco's Mediterranean and semi-arid climate gradient and similar dominant crops, it lacks the dense, parcel-level, field-verified irrigation ground-truth data available for Morocco - making it representative of the data-scarce settings where scalable mapping approaches are most needed. Reliable irrigated area information is needed to guide irrigation investment and monitor food security, yet ground truth data are extremely scarce and landscape heterogeneity makes remote sensing-based mapping particularly challenging (Kerner et al., 2020). Developing approaches that can be trained in data-rich environments and transferred effectively to agroclimatically similar regions would therefore represent a significant advance for agricultural monitoring across North Africa and beyond.

1.2 State of the Art in Irrigated Area Mapping

Remote sensing has long been the primary approach for mapping irrigated cropland at regional and global scales. Early global products such as the Global Map of Irrigation Areas (GMIA) by Siebert et al. (2005) and the Global Irrigated Area Map (GIAM) by Thenkabail et al. (2009) established the feasibility of satellite-based inventory, but these products operated at coarse resolutions of 5 arc-minutes (~9 km) to 10 km, which are entirely unsuitable for identifying the small, fragmented fields that dominate East African agricultural landscapes. A significant step forward came with the Landsat-derived Global Rainfed and Irrigated-Cropland Product at 30 m resolution (LGRIP30; Teluguntla et al., 2023), produced using advanced machine learning algorithms, achieving an overall accuracy of 86.5%. LGRIP30 estimates a total global net irrigated area of 0.71 Gha from 1.80 Gha of total cropland - an irrigation proportion of approximately 39%, notably higher than earlier estimates - and represents the highest-resolution global irrigated area product

currently available from a peer-reviewed, open-access source. More recently, the Global Maximum Irrigation Extent (GMIE) at 30 m resolution by Wu et al. (2023) represents further progress through a drought-stress threshold approach; however, this product reported an irrigation proportion of only 1.1% for western Africa - the lowest of any global region - and its threshold-based methodology requires locally calibrated parameters that perform poorly under the heterogeneous, frequently cloud-covered conditions of tropical East Africa. A further limitation shared by all these products is temporal: they provide static snapshots or decade-maximum extents, not the annual time series that irrigation monitoring and policy evaluation require (Wu et al., 2023; Siebert et al., 2005; Teluguntla et al., 2023).

Regional African studies have consistently reinforced these limitations. Chandrasekharan et al. (2021) mapped irrigated and rainfed agriculture across Ethiopia and found that global products overestimate irrigated areas by conflating wetlands and evergreen vegetation with irrigated cropland, a problem driven by coarse pixel resolution and the application of generalised classification rules developed for entirely different landscapes. Gumma et al. (2011) mapped irrigated areas in Ghana using fused Landsat and MODIS data and found that inland valleys, shallow dug-wells, and fragmented smallholder plots - the dominant forms of irrigation in the region - are systematically missed by coarse-resolution approaches, with remote sensing still overestimating areas by 20-57% compared to national statistics. Both studies point to the same fundamental problem: irrigation mapping methods developed and validated in large, contiguous, data-rich agricultural systems in Asia or North America do not transfer reliably to Africa's heterogeneous, data-scarce, smallholder-dominated landscapes.

Machine learning classifiers applied to multi-temporal Sentinel-1 SAR and Sentinel-2 time series represent the current state of the art for fine-resolution irrigation mapping. Although these data-driven methods represent a substantial advance over rule-based classifiers, they continue to face three fundamental challenges: strong dependence on manually engineered spectral and temporal features, limited robustness to domain shift when models trained in one region are applied to another, and difficulty integrating observations from multiple sensors with different spatial resolutions and revisit frequencies (Yang et al., 2026). These limitations are especially severe for irrigated cropland identification, which depends on crop phenology, irrigation timing, and field management practices that vary substantially across regions and irrigation system types.

1.3 Geospatial Foundation Models: A New Paradigm

The emergence of geospatial foundation models represents a qualitative shift in the methodological landscape. Foundation models are large neural networks pre-trained via self-supervised learning on massive unlabelled datasets, which can then be fine-tuned for specific downstream tasks using only a small number of labeled examples (Tseng et al., 2023). Applied to Earth observation, they learn compact yet semantically rich spatiotemporal representations directly from raw satellite imagery - without requiring manual feature engineering - and can in principle transfer these representations across geographic regions, sensor modalities, and application domains (Szwarcman et al., 2024; Yang et al., 2026).

Several geospatial foundation models have been developed in recent years, each differing in architecture, pre-training data, and sensor integration strategy. Prithvi-EO-2.0 (NASA × IBM) is a multi-temporal Vision Transformer trained on 4.2 million global time-series samples from the Harmonised Landsat-Sentinel-2 archive at 30 m resolution, incorporating explicit temporal and location embeddings that capture seasonal vegetation dynamics relevant to crop mapping (Szwarcman et al., 2024). AgriFM is the first foundation model designed exclusively for agricultural mapping, using a modified Video Swin Transformer pre-trained on over 25 million samples from MODIS, Landsat, and Sentinel-2, demonstrating consistent superiority over general-purpose remote sensing foundation models across crop identification and field boundary benchmarks (Li et al., 2025). TerraMind (IBM × ESA) is the first any-to-any generative multimodal Earth observation foundation model, pre-trained on nine geospatial modalities - including Sentinel-1, Sentinel-2, and DEM - across 9 million spatiotemporally aligned samples, with

a distinctive Thinking-in-Modalities capability that generates auxiliary modalities during inference to improve task performance (Jakubik et al., 2025). The Clay Foundation Model is an open-source 632-million parameter Vision Transformer trained on 70 million images from multiple satellite platforms including Sentinel-1 SAR and Sentinel-2, using a masked autoencoder framework augmented with DINO-based knowledge distillation to produce embeddings encoding spatial, temporal, and location information simultaneously (Clay, 2024). SatMAE is a masked autoencoder pre-trained on multi-temporal and multi-spectral satellite imagery from the fMoW-Sentinel dataset, demonstrating strong transfer learning capabilities for land cover classification tasks through its temporal group positional encoding (Cong et al., 2022). Scale-MAE extends masked autoencoders to handle variable ground sampling distances, encoding scale information into position embeddings to handle multi-resolution imagery and showing strong performance across spatial scales (Reed et al., 2023). Presto is a lightweight pixel-timeseries transformer pre-trained on globally distributed multi-sensor observations including Sentinel-1, Sentinel-2, ERA5 meteorological data, and DEM, specifically designed for label-scarce agricultural settings and used operationally in the WorldCereal global crop and irrigation mapping system (Tseng et al., 2023; Van Tricht et al., 2023).

Although geospatial foundation models encode rich representations from large-scale pre-training, direct application to domain-specific tasks without adaptation typically yields suboptimal performance because the models have not been exposed to the specific spectral, temporal, and geographic patterns of the target region or task. Several adaptation strategies exist - ranging from a frozen encoder approach where only a lightweight decoder is trained, to full fine-tuning of all model parameters. Among parameter-efficient approaches, Low-Rank Adaptation (LoRA) has emerged as particularly effective, introducing trainable low-rank matrices into selected transformer layers while keeping the original encoder weights frozen, thereby reducing memory requirements and mitigating catastrophic forgetting while improving generalisation to unseen regions (Marti-Escofet et al., 2025). Recent studies have also shown that combining embeddings from multiple foundation models can improve downstream Earth observation tasks by leveraging complementary information across architectures (van der Plas et al., 2026).

Despite this growing landscape, it remains unclear which modelling paradigm is most effective for irrigated cropland mapping specifically, and whether learned representations transfer reliably across the diverse agroclimatic and agricultural conditions of Africa. This study evaluates four models - Prithvi-EO-2.0, AgriFM, TerraMind, and Clay - selected on the basis of three criteria. First, all four provide publicly available pre-trained weights enabling fine-tuning, unlike models such as AlphaEarth Foundation whose weights are not released. Second, all four support multi-temporal Sentinel-1 and Sentinel-2 input, which are the primary satellite data sources used in this study. Third, the four models together represent the full spectrum of current GFM architectures - MAE-based vision transformers (Prithvi-EO-2.0, Clay), an agriculture-specific temporal transformer (AgriFM), and an any-to-any generative multimodal model (TerraMind) - enabling a systematic assessment of whether architectural differences and pre-training choices drive meaningful performance differences for irrigation mapping.

1.4 The Research Gap

Despite significant advances in remote sensing-based irrigation mapping, important scientific and practical gaps remain.

First, no annual irrigated cropland maps exist for Morocco or for other North African countries. Existing global products such as the GMIE (Wu et al., 2023) provide only static decade-maximum extents, not the year-by-year maps that governments need to monitor irrigation expansion, evaluate policy outcomes, and track water use over time. Producing such a time series requires a model that is not only accurate at a single point in time but also temporally consistent - able to detect irrigated areas reliably across different years, and climatic conditions. Whether fine-tuned foundation models can achieve this temporal stability, and whether that stability holds when the model is applied to a different country altogether, has never been tested.

Second, despite the growing momentum behind geospatial foundation models, none has been evaluated or fine-tuned for irrigated cropland mapping. No study has tested whether foundation models can produce accurate irrigated cropland maps in Africa's diverse agro-hydrological landscapes. Morocco presents a particularly demanding test case: its major agricultural regions span semi-arid to sub-humid climates, support a diverse range of crop types from seasonal cereals to perennial tree crops, and employ irrigation systems ranging from traditional flood irrigation to advanced drip and pivot methods (Ouassanouan et al., 2026).
Even when multiple foundation models exist, it is unknown which one is best suited for irrigated cropland mapping, and whether the best-performing model transfers spatially to data-scarce regions. Each of the four candidate models - Prithvi-EO-2.0, AgriFM, TerraMind, and Clay - differs in architecture, training data, and sensor integration strategy, and no study has systematically compared them for this specific task. Furthermore, while recent studies suggest that different foundation models may learn complementary representations, the potential of multi-model embedding fusion for irrigated cropland mapping remains unexplored. Whether a model trained on Morocco's diverse Mediterranean and semi-arid landscapes generalises reliably to Tunisia - which shares the same agroclimatic zone but lacks the dense parcel-level ground truth needed for local model development - is unknown. If foundation models trained in Morocco can generate useful irrigated area maps for data-scarce countries without additional labeled data, they offer a scalable and practical pathway to irrigation monitoring across the continent.

This research will address these gaps: four foundation models will be fine-tuned and compared on the Morocco ground-truth dataset, the best-performing model will be identified through systematic benchmarking, and annual irrigated cropland maps for the period 2021-2025 will be produced for Morocco. Both the spatial and temporal transferability of the selected model will then be evaluated in Tunisia - an agroclimatically similar North African country where no dense field-verified irrigation ground truth currently exists.

2. Objectives and Research Questions

2.1 Main Objective

To compare the performance and transferability of geospatial foundation models for irrigated cropland mapping.

2.2 Sub-Objectives and Related Research Questions

Objective 1 - To compare the performance of four geospatial foundation models and adaptation strategies for irrigated cropland mapping in Morocco

- RQ1: What classification accuracy does each foundation model achieve for irrigated cropland mapping across Morocco's diverse agricultural regions, and how does performance compare to a conventional Sentinel-based Random Forest classifier?
- RQ2: To what extent does PEFT (parameter-efficient fine-tuning) improve irrigated cropland classification performance compared with a frozen encoder strategy?

Objective 2 - To evaluate the temporal and spatial transferability of the best-performing model through annual irrigated cropland mapping in Morocco (2021-2025) and zero-shot transfer to Tunisia.

- RQ3: How temporally consistent is the best-performing model when applied across multiple years, and does the resulting annual map series for Morocco reflect known irrigation expansion trends reported in national statistics?
- RQ4: When the best-performing Morocco-trained model is applied without retraining to Tunisia, how does irrigation mapping accuracy degrade spatially relative to within-Morocco performance, and does the model maintain temporal consistency when producing annual maps in the transfer region?

Objective 3 (Optional Extension): To investigate whether combining representations from multiple geospatial foundation models through embedding fusion improves irrigated cropland mapping performance.

- RQ5: Can feature-level fusion of embeddings extracted from the best-performing geospatial foundation models improve irrigated cropland classification performance compared with individual foundation models?

Note: This objective is considered an optional extension and will only be pursued if time and computational resources permit following completion of the core research objectives.

3. Study Area and Datasets

3.1 Primary Study Area - Morocco

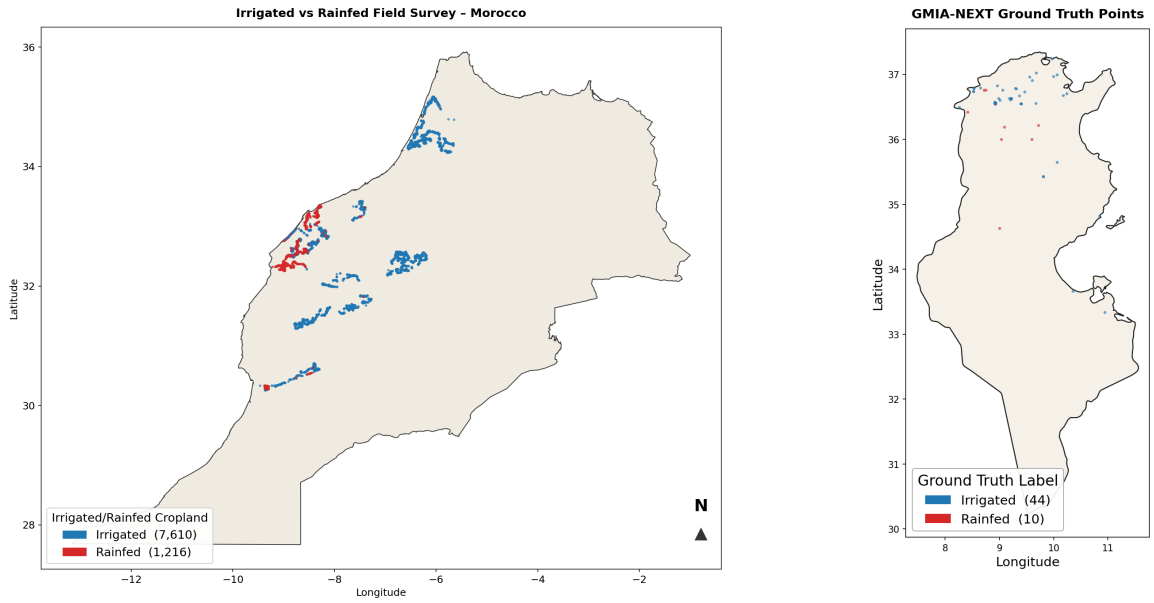
Morocco spans a wide climatic gradient from humid in the north to arid in the south, making it one of the most agriculturally diverse countries in Africa. This study focuses on major agricultural regions that together capture this diversity: El Gharb (humid, surface water irrigation, rice and vegetables), Tadla (semi-arid to sub-humid, large-scale irrigation, sugar beets and cereals), Doukkala (Mediterranean, mixed irrigated and rainfed systems), El Haouz (semi-arid, groundwater irrigation, olives and citrus), and Souss (arid, advanced drip systems, high-value export crops) (Ouassanouan et al., 2026). The range of climates, water sources, and irrigation systems across these regions and the availability of field-verified ground truth data make Morocco a demanding, practical, and representative test environment for foundation model evaluation.

3.2 Transfer Study Area - Tunisia

To evaluate the spatial transferability of geospatial foundation models to a data-scarce environment, the best-performing Morocco-trained model will be applied to Tunisia without additional retraining. Tunisia was selected as the transfer study area because it shares Morocco's Mediterranean and semi-arid climate gradient, similar dominant crops and comparable irrigation system types - making it an agroclimatically appropriate transfer target. However, unlike Morocco, Tunisia lacks the dense parcel-level field-verified irrigation ground-truth data needed for local model development, representing the data-scarce conditions under which scalable foundation model transfer is most practically relevant. For accuracy assessment in Tunisia, ground-truth points from the GMIA-NEXT global irrigated area dataset (Kebede et al., 2026) will be used.

3.3 Datasets

Ground-truth data:
Two labeled datasets will be used in this study. For Morocco, the primary resource is the openly accessible crop and irrigation ground-truth dataset of 8,826 georeferenced parcels collected across major agricultural area, each recording crop type, irrigation status, and irrigation system type (Ouassanouan et al., 2026), constituting the core training and validation resource. For Tunisia, ground-truth points will be extracted from the GMIA-NEXT global irrigated and rainfed cropland dataset (Kebede et al., 2026), which provides 380,696 globally distributed binary irrigated and rainfed labels corresponding to the 2023/24 growing season. Filtering to Tunisia yields 54 validation points - 44 irrigated and 10 rainfed - concentrated primarily in northern region.



(a) (b)

Figure 1: Location and spatial distribution of ground-truth data used in this study. (a) Morocco: distribution of 8,826 field-surveyed parcels across Morocco's major agricultural regions, showing 7,610 irrigated and 1,216 rainfed parcels (Ouassanouan et al., 2026). (b) Tunisia: distribution of 54 GMIA-NEXT validation points - 44 irrigated and 10 rainfed - concentrated primarily in northern Tunisia (Kebede et al., 2026).

Satellite imagery: Annual Sentinel-2 multispectral (10 m) and Sentinel-1 SAR (VV/VH) composites will be prepared for 2021-2025 using STAC, xarray, providing cloud-independent multi-temporal inputs for model fine-tuning and annual map production.

Validation and comparison datasets: Classification outputs will be compared against the GMIE (Wu et al., 2023), LGRIP30 (Teluguntla et al., 2023), FAO WaPOR, and FAO AQUASTAT national irrigated area statistics to contextualise model performance relative to existing global products.

4. Methods

The proposed methods consists of four main phases: (1) data preparation, (2) model training and evaluation, (3) annual irrigated cropland mapping, and (4) transferability assessment. First, Sentinel-1 and Sentinel-2 imagery and parcel-level reference data will be prepared for model development. Second, multiple geospatial foundation models will be evaluated using frozen encoder and parameter-efficient fine-tuning strategies. An optional multi-model fusion experiment will also be conducted to investigate whether combining representations from multiple foundation models improves classification performance. Third, the best-performing model will be used to generate annual irrigated cropland maps for Morocco between 2021 and 2025. Finally, the selected model will be transferred to Tunisia to assess its ability to generalise to a data-scarce agricultural environment. Figure 2 illustrates the overall methodological framework of the study.
As a conventional machine-learning baseline, a Random Forest classifier will be trained using Sentinel-derived spectral and radar features, including vegetation and moisture indices together with Sentinel-1 backscatter information. This baseline will provide a benchmark against which the performance of geospatial foundation models can be assessed.

4.1 Data Preparation

Annual multi-temporal image stacks will be generated from Sentinel-1 Synthetic Aperture Radar (SAR) and Sentinel-2 multispectral imagery for the period 2021-2025. Sentinel-2 provides optical observations of vegetation condition and crop development, while

Sentinel-1 provides cloud-independent information on crop structure and surface moisture. To preserve temporal information relevant to irrigation dynamics, annual image stacks will be constructed from monthly Sentinel-1 and Sentinel-2 composites, enabling the models to capture vegetation and moisture variations throughout the agricultural year. A 10 m cropland mask will be applied prior to classification to restrict the analysis to agricultural areas and reduce confusion with non-cropland classes such as forests, urban areas, water bodies, and bare land, so that the classification task focuses exclusively on distinguishing irrigated from rainfed cropland. Since the precision of any single global cropland mask varies by region and may be lower in areas where training data are sparse, multiple cropland datasets will be compared for both Morocco and Tunisia, including WorldCereal (Van Tricht et al., 2023), ESA WorldCover 2021 (Zanaga et al., 2022),and the GLAD global cropland extent map (Potapov et al., 2021). Where individual products show disagreement or apparent omissions of known agricultural areas, a consensus mask combining pixels classified as cropland by at one product will be used to maximise cropland coverage and reduce mask-related omission errors prior to classification.

The primary reference dataset will be the open-access Moroccan agricultural parcel dataset developed by Ouassanouan et al. (2026), which contains approximately 8,826 georeferenced parcels. Because the field observations correspond to the 2024 agricultural season, model training and benchmarking will be conducted using the 2024 annual image stack. Samples will be divided into training (70%) and testing (30%) subsets using stratified sampling to ensure proportional representation of irrigation classes and study regions. A validation subset will be extracted from the training data for model selection and hyperparameter tuning.

For Tunisia, ground-truth points will be extracted from the GMIA-NEXT global irrigated and rainfed cropland dataset (Kebede et al., 2026), corresponding to the 2023/24 growing season. Filtering to Tunisia yields 54 validation points - 44 irrigated and 10 rainfed - concentrated primarily in northern Tunisi

4.2 Foundation Model Adaptation and Comparative Evaluation

Four geospatial foundation models will be evaluated: Prithvi-EO-2.0, AgriFM, TerraMind, and Clay. A detailed comparison of the evaluated models is provided in Appendix A. Three adaptation strategies will be investigated: a Frozen Encoder strategy, Parameter-Efficient Fine-Tuning (PEFT), and Full Fine-Tuning.

4.2.1 Frozen Encoder Strategy

The pre-trained encoder weights will remain fixed, and only the decoder and classification head will be trained. This experiment will assess the usefulness of the representations learned during large-scale pre-training without further adaptation and will provide a computationally efficient baseline against which fine-tuning approaches can be evaluated.

4.2.2 Parameter-Efficient Fine-Tuning (PEFT)

Recent studies have shown that Parameter-Efficient Fine-Tuning (PEFT) methods can achieve performance comparable to, and in some cases better than, full fine-tuning while requiring substantially fewer trainable parameters and computational resources (Marti-Escofet et al., 2025). Several PEFT approaches have been proposed, including Low-Rank Adaptation (LoRA), Quantized LoRA (QLoRA), Weight-Decomposed Low-Rank Adaptation (DoRA), Adaptive LoRA (AdaLoRA), Visual Prompt Tuning (VPT), ViT Adapters, and BitFit. These approaches adapt foundation models by updating only a small subset of parameters while preserving most of the pre-trained model weights. A comparative overview of these fine-tuning strategies is provided in Appendix B.

Among these methods, LoRA has emerged as one of the most effective and widely adopted approaches for geospatial foundation models. LoRA introduces trainable low-rank matrices into selected transformer layers while keeping the original model weights frozen. This enables task-specific adaptation while preserving knowledge learned during large-scale pre-training, reducing memory requirements, and mitigating the risk of catastrophic forgetting. Furthermore, LoRA has been shown to better preserve geographic representations and improve generalisation to unseen regions, making it particularly suitable for transferability studies (Marti-Escofet et al., 2025).

Consequently, LoRA will be adopted as the primary PEFT strategy in this research. Low-rank trainable adapters will be inserted into selected transformer layers while the original encoder weights remain frozen. During training, only the LoRA parameters, decoder, and classification head will be updated. The performance of LoRA-based adaptation will be compared with the Frozen Encoder strategy to assess the benefits of task-specific adaptation for irrigated cropland mapping.

4.2.3 Full Fine-Tuning (Optional)

Full fine-tuning updates all model parameters during training and therefore provides the maximum degree of task-specific adaptation. However, this approach requires substantially greater computational resources, GPU memory, and training time compared with PEFT methods. If time and computational resources permit, full fine-tuning will be conducted for selected foundation models, preferably Prithvi-EO-2.0 or Clay, as these models have well-documented fine-tuning workflows and comparatively manageable model sizes. This experiment will assess whether the additional computational cost of full fine-tuning leads to meaningful performance improvements over LoRA-based adaptation and Frozen Encoder strategies. Otherwise, the core comparison will focus on Frozen Encoder and Parameter-Efficient Fine-Tuning adaptation.

4.2.4 Comparative Evaluation

Model performance will be assessed using spatial cross-validation to minimise spatial leakage between training and testing data. Rather than randomly splitting samples, entire regions will be withheld during testing to better evaluate model generalisation across different agricultural environments.

Performance will be evaluated using standard classification metrics, including Overall Accuracy, Precision, Recall, and F1 score. Comparisons will be conducted between:

1. Random Forest and geospatial foundation models
2. Frozen Encoder, PEFT strategies (primarily LoRA, and other PEFT methods where applicable), and Full Fine-Tuning (where applicable)
3. Different geospatial foundation models
4. Individual models and fused multi-model representations (optional).

The best-performing model and adaptation strategy combination will be selected based on the highest mean F1-score across the five cross-validation folds, with standard deviation used as an indicator of consistency across regions - a lower standard deviation indicating more stable generalisation across Morocco's diverse agricultural conditions.

4.3 Multi-Model Fusion

Recent studies have shown that Earth observation foundation models may learn complementary representations due to differences in architecture, pre-training objectives, and training datasets (van der Plas et al., 2026). Therefore, an exploratory fusion experiment may be conducted to investigate whether combining representations from multiple geospatial foundation models improves irrigated cropland mapping performance. Following the comparative evaluation, embeddings will be extracted from the best-performing foundation models and combined using feature-level fusion through embedding concatenation. The fused representations will then be used as input to a lightweight classifier. Classification performance will be evaluated using the same metrics applied to individual models, including Overall Accuracy (OA), Precision, Recall, and F1-score. The performance of fused model representations will be compared against the best-performing individual foundation model to assess whether complementary information captured by different models leads to improved irrigated cropland classification. Due to additional computational requirements, this experiment will be considered an optional extension and will only be undertaken if time and computational resources permit.

4.4 Annual Irrigated Cropland Mapping (2021-2025)

The best-performing model identified during benchmarking will be applied to annual Sentinel-1 and Sentinel-2 image stacks for all years between 2021 and 2025. Classification will be restricted to cropland mask. This approach will produce a series of annual irrigated cropland maps describing the spatial distribution of irrigation across Morocco over the five-

year study period. Annual irrigated area estimates will be compared with existing irrigation products, including the Global Maximum Irrigation Extent (GMIE), the Landsat-derived Global Rainfed and Irrigated Cropland Product (LGRIP30), and FAO WaPOR datasets. Where available, comparisons will also be made with national agricultural statistics and reported irrigation development trends. These analyses will provide an assessment of the temporal consistency and practical applicability of foundation-model-derived irrigation maps.

4.5 Cross-Region Transferability Assessment

To evaluate spatial transferability, the best-performing Morocco-trained model will be applied directly to Tunisia without additional retraining. This zero-shot transfer experiment will assess whether representations learned from Morocco's agricultural landscape can generalise to Tunisia — a region sharing a similar Mediterranean and semi-arid climate gradient but lacking the dense field-verified ground truth needed for local model development.

The same preprocessing workflow used in Morocco will be applied to Tunisia, including annual Sentinel-1 and Sentinel-2 image stacks and the cropland mask. Annual irrigated cropland maps will be generated for the period 2021-2025.
Where reference information is available, model outputs will be compared with existing irrigation datasets and agricultural monitoring products, including FAO WaPOR, GMIE, and LGRIP30. Performance degradation relative to Morocco will be quantified to assess the robustness of foundation-model representations under geographic transfer. The results will provide insight into the potential of geospatial foundation models for scalable irrigated cropland mapping in data-scarce regions in Tunisia.

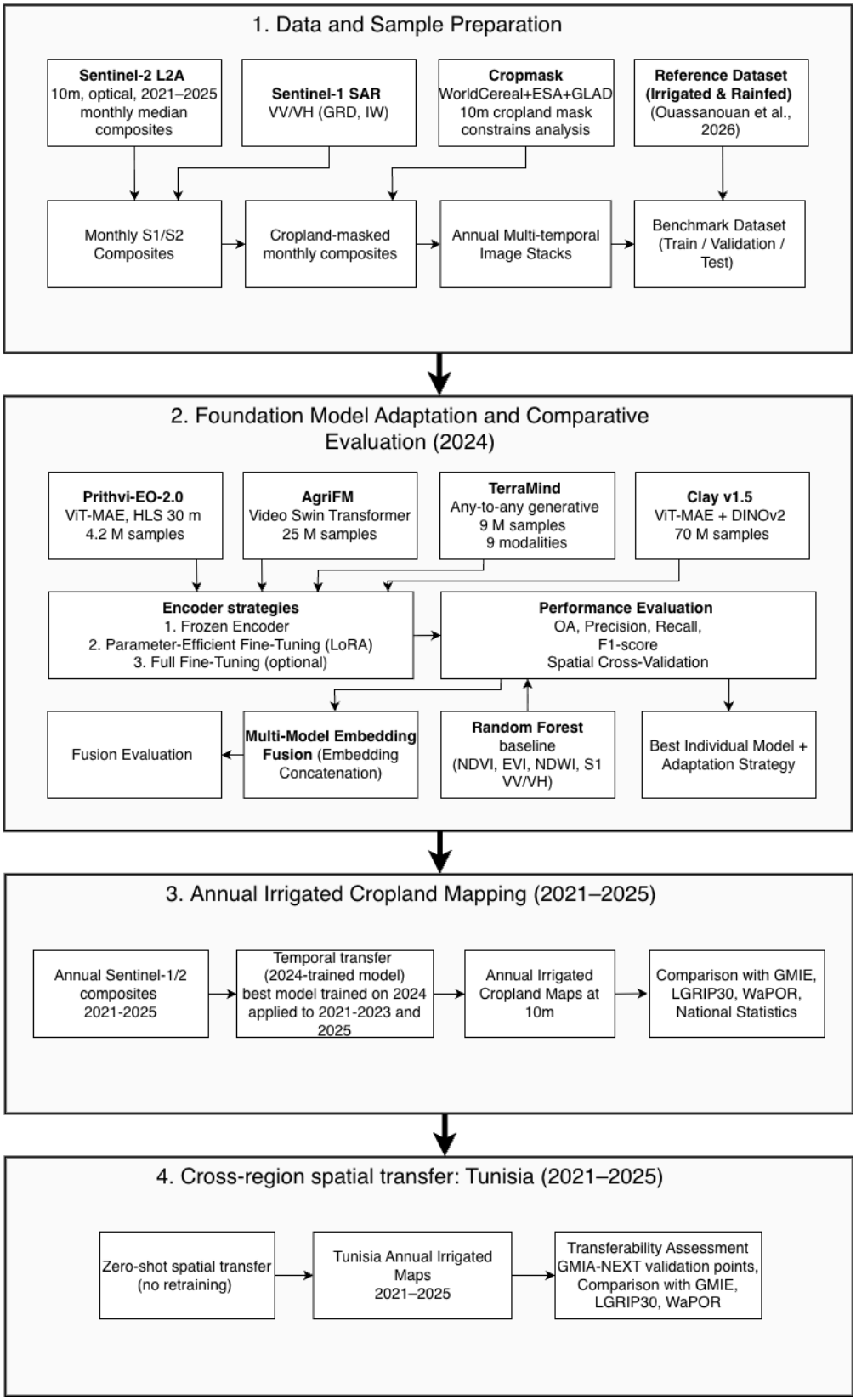


Figure 2: Methodological workflow for evaluating geospatial foundation models for irrigated cropland mapping in Morocco and assessing cross-region transferability to Tunisia.

5. Expected Outcomes and Scientific Contribution

5.1 Expected Outcomes

The proposed research is expected to produce:

1. A comprehensive evaluation of four geospatial foundation models (Prithvi-EO-2.0, AgriFM, TerraMind, and Clay) for irrigated cropland mapping in Morocco.
2. A comparative assessment of Frozen Encoder and Parameter-Efficient Fine-Tuning (PEFT) strategies, providing insights into the effectiveness of task-specific adaptation for irrigation mapping.
3. Annual irrigated cropland maps for Morocco at 10 m spatial resolution for the period 2021-2025.
4. An evaluation of the temporal consistency of foundation-model predictions through comparison with existing irrigation products and national statistics.
5. An assessment of the spatial transferability of the best-performing model through zero-shot transfer from Morocco to Tunisia.

5.2 Scientific Contribution

This research will contribute to the growing field of GeoAI and Earth Observation Foundation Models in several ways.

- First, it will provide one of the first systematic evaluations of geospatial foundation models for irrigated cropland mapping in Africa. While foundation models have recently demonstrated strong performance in land-cover mapping and crop classification, their potential for irrigation mapping remains largely unexplored.
- Second, the study will investigate whether parameter-efficient fine-tuning offers advantages over frozen foundation model representations for irrigated cropland classification. This contributes to the emerging literature on efficient adaptation strategies for Earth observation foundation models.
- Third, the research will provide empirical evidence regarding the spatial transferability of foundation models across African agricultural systems. Understanding whether models trained in relatively data-rich environments can generalise to data-scarce regions is essential for developing scalable agricultural monitoring systems.
- Finally, the study will generate annual irrigated cropland maps for Morocco and evaluate their consistency against existing irrigation products, providing insights into the suitability of foundation models for operational irrigation monitoring and water-resource management.

6. Risks and Mitigation Strategies

Although PEFT substantially reduces the number of trainable parameters compared with full fine-tuning, adapting large geospatial foundation models may still require significant computational resources, particularly for models with large encoder architectures and multi-temporal inputs. Computational limitations may therefore constrain the number of experiments, hyperparameter configurations, or foundation models that can be evaluated. To mitigate this risk, the study adopts a parameter-efficient fine-tuning strategy, prioritises models with publicly available pre-trained weights, and includes a Frozen Encoder approach as a computationally lightweight baseline. If hardware limitations are encountered, experiments will be prioritised based on model performance and feasibility. Full Fine-Tuning will be considered only for selected foundation models and will be conducted if sufficient hardware resources are available. If computational constraints are encountered, experiments will be prioritised according to scientific relevance and feasibility, with the core comparison focusing on Frozen Encoder and LoRA-based adaptation strategies.

7. Work Plan and Time Schedule

Activity	Month
Literature Review and Proposal Finalization	June - July 2026
Summer Vacation (Travel to India)	July-August 2026
Data Acquisition, Preprocessing, Benchmark Dataset Construction and Random Forest Baseline Development	September 2026
Foundation Model Implementation and Frozen Encoder Experiments	September - October 2026
Parameter-Efficient Fine-Tuning Experiments	November - December 2026
Full Fine-Tuning (Optional), and Multi-Model Fusion Experiments	December 2026 - January 2027
Comparative Evaluation of the Models	February 2027
Annual Irrigated Cropland Mapping (2021-2025) and Spatial Transferability Assessment	March 2027
Results Analysis and Interpretation	April 2027
Thesis Writing	April - May 2027
Thesis Revision and Finalization	May 2027
Thesis Submission and Defence Preparation	June 2027

Note: During the period September-November 2026, an internship is planned alongside thesis activities. The preferred option is a three-month internship, potentially involving approximately two months on-site followed by remote work from Enschede. Potential host organisations include Expo City Dubai or the World Bank Group. The internship schedule may be adjusted depending on thesis progress and host organisation requirements.

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Statement on the Use of AI Tools

AI tools were used in the preparation of this proposal. Specifically, Claude (Anthropic, claude.ai) was used alongside Scopus and Google Scholar to support literature discovery, and at selected points to assist with summarising relevant papers on irrigated area mapping, geospatial foundation models, and checking APA 7th citation formatting and reference list completeness. All core arguments, identification of research gaps, research framing, and intellectual contributions are my own. The final text was critically reviewed, substantially revised, and verified against the cited sources by me. This declaration is made in accordance with the University of Twente guidelines on the use of AI in academic work.

Appendices

Appendix A. Geospatial Foundation Models Evaluated in this Study

Model	Developer	Architecture	Pre-training data	Parameters	Input sensors
Prithvi-EO-2.0	NASA × IBM	ViT-MAE (3D spatiotemporal)	HLS 30 m - 4.2M global time-series samples	100M / 300M / 600M	Landsat + Sentinel-2
AgriFM	Northwest A&F University	Video Swin Transformer (multi-source temporal)	MODIS + Landsat + Sentinel-2 - 25M samples	Not disclosed	MODIS, Landsat, Sentinel-2
TerraMind	IBM × ESA	Dual-scale Transformer - any-to-any generative	TerraMesh - 9M spatiotemporal	Not disclosed	Sentinel-1, Sentinel-2, DEM

			samples, 9 modalities		+ 6 more
Clay v1.5	Development Seed	ViT-MAE + DINOv2 - any sensor/resolution	S1 + S2 + Landsat 8/9 + NAIP + MODIS - 70M chips	632M	Any sensor, any number of bands

Table 1. Comparison of the four geospatial foundation models evaluated in this study. Prithvi-EO-2.0, AgriFM, Clay, and TerraMind provide open model weights enabling fine-tuning. All differ in architecture, pre-training data volume, and sensor fusion strategy, motivating a systematic comparison for irrigated cropland mapping.

Appendix B. Parameter-Efficient Fine-Tuning (PEFT) Strategies

Strategy	Core Idea	Advantages	Limitations
LoRA (Low-Rank Adaptation)	Introduces trainable low-rank matrices into transformer layers while keeping original weights frozen.	Strong performance, memory efficient, preserves pre-trained knowledge, widely adopted.	Requires tuning of rank and adapter parameters.
QLoRA (Quantized LoRA)	Combines LoRA with low-bit quantisation of model weights.	Further reduces GPU memory requirements.	Quantisation may slightly affect performance.
DoRA (Weight-Decomposed LoRA)	Separates weight magnitude and direction during adaptation.	Can improve adaptation quality compared with standard LoRA.	More complex implementation and training.
AdaLoRA (Adaptive LoRA)	Dynamically allocates adaptation parameters to important layers during training.	Efficient use of trainable parameters.	Additional hyperparameter tuning required.
Visual Prompt Tuning (VPT)	Learns a small set of trainable prompt tokens while keeping the backbone frozen.	Extremely parameter efficient.	May provide lower performance gains than LoRA.
ViT Adapter	Inserts lightweight adapter modules between transformer layers.	Good balance between adaptation and efficiency.	Requires architectural modifications.
BitFit	Updates only bias parameters while keeping all other weights fixed.	Simplest and most computationally efficient approach.	Limited adaptation capacity.
Full Fine-Tuning	Updates all model parameters during training.	Maximum adaptation flexibility.	High memory usage, longer training time, greater risk of overfitting.

Table 2. Comparison of Parameter-Efficient Fine-Tuning (PEFT) Strategies

Appendix C. Data Sources

Data Source	Provider / Source	Data Type	Purpose in Study	Rights / Access
Sentinel-1 GRD	European Space Agency (ESA)	SAR imagery	Multi-temporal radar observations for irrigated cropland mapping	Open access (Copernicus Data Policy)
Sentinel-2 Level-2A	European Space Agency (ESA)	Optical multispectral imagery	Vegetation and crop condition monitoring	Open access (Copernicus Data Policy)
WorldCereal Cropland Mask	WorldCereal Consortium	Cropland mask	Consensus cropland mask — restrict analysis to agricultural land	Open access
ESA WorldCover 2021	Zanaga et al. (2022)	(2022)10 m global land cover	Consensus cropland mask — restrict analysis to agricultural land	Open access (CC-BY 4.0)
GLAD Global Cropland	Potapov et al. (2021)	Global cropland extent map	Consensus cropland mask — restrict analysis to agricultural land	Open access
Morocco Agricultural Parcel Dataset	Ouassanouan et al. (2026)	Parcel polygons with irrigation labels	Training, validation, and testing data	Open-access dataset; citation required
GMIA-NEXT Ground Truth Points	Kebede et al. (2026)	Global irrigated/rainfed point labels	Validation data for Tunisia transfer experiment	Open access (Zenodo)
GMIE (Global Maximum Irrigation Extent)	IWMI and partners	Irrigation map	Validation and comparison of annual irrigation estimates	Open access; citation required
LGRIP30	Teluguntla et al. (2023)	Global irrigated/rainfed cropland map	Validation and comparison of annual irrigation estimates	Open access; citation required
FAO WaPOR	Food and Agriculture Organization (FAO)	Agricultural monitoring and evapotranspiration products	Validation and interpretation of irrigation patterns	Open access (registration required)
FAO AQUASTAT	Food and Agriculture Organization (FAO)	Irrigation and water-use statistics	Contextual comparison of irrigated area estimates	Open access

Table 3. Summary of Data Sources Used in the Study

Appendix D. Estimated Hardware and Computational Resources

Although parameter-efficient fine-tuning substantially reduces memory and computational requirements compared with full fine-tuning, training and benchmarking multiple geospatial foundation models may still require access to high-memory GPUs and large storage capacity for Earth observation data processing.

Resource	Recommended Configuration
CPU	32+ cores
System Memory (RAM)	64-128 GB
GPU	NVIDIA A100 / H100 / RTX 6000 Ada with 40-80 GB VRAM
GPU Memory	40-80 GB
Storage (SSD)	1-2 TB
Operating System	Linux (Ubuntu 22.04 LTS or equivalent)

Table 4. Estimated Computational Resources for Model Training and Evaluation